

Richard Alexander

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EDUCATION

Sep 2025 – Jun 2027	MSc Human-Computer Interaction and Design (HCID) Double-degree programme, minor in Innovation & Entrepreneurship	<i>EIT Digital Master School</i>
Aug 2026 – Jun 2027	KTH Royal Institute of Technology , Stockholm (<i>exit year, current</i>)	
Sep 2025 – Jul 2026	University of Twente , Enschede (<i>entry year, MSc Interaction Technology, completed</i>)	
Sep 2019 – May 2023	BSc Computer Games Development, Computer Science University of Limerick, QCA 3.37	<i>Limerick</i>

EMPLOYMENT HISTORY

Apr 2025 – Sep 2025	Full-stack Developer, IBM - Supported IBM API Connect, IBM's API management platform (Gateway, Portal, Management and Analytics components, deployed via Cloud Pak across on-prem and SaaS), resolving customer-reported integration issues. - Handled tickets specifically around integrating webMethods, Software AG's API management product, into API Connect following IBM's acquisition of webMethods, diagnosing cross-platform compatibility issues. - Worked hands-on across the platform's full stack: Node.js, React, NGINX, PostgreSQL, OpenAPI Specification, OAuth2 and GraphQL. - Operated within Kubernetes/Docker and Red Hat OpenShift-based deployments, using Helm for packaging and Jenkins for CI/CD, to investigate and resolve platform issues.	<i>Cork</i>
Feb 2024 – Apr 2025	Software Developer, General Motors - Completed an intensive 8-month self-directed training and assessment program in mainframe technologies (COBOL, JCL, REXX, Easytrieve, PL/I). - Contributed to mainframe modernization projects, maintaining and improving legacy production systems. - Used Azure DevOps for pipeline builds and Git for version control across collaborative team workflows. - Worked closely with senior developers and fellow new hires, picking up new tools and languages quickly as projects required.	<i>Limerick</i>
Jan 2022 – Aug 2022	Software Engineer, Ericsson - Worked in an agile team setting up and maintaining microservices on Ubuntu/Linux infrastructure. - Used Git and Gerrit for code review and version control, and Docker/Kubernetes for containerised deployment. - Picked up JavaScript on the job with support from senior engineers; tracked work in Jira.	<i>Athlone</i>
Sep 2020 – Jun 2021	Teaching Assistant, University of Limerick - Led peer-supported learning groups for 1st-year Java Programming students at the ICT Learning Centre, running weekly sessions through pandemic lockdowns. - Applied problem-based learning strategies to strengthen students' understanding of module content. - Taught secondary-school students at UL Cybercamps across Basic Web Design (HTML/CSS), Java with Greenfoot, and visual programming with Processing.	<i>Limerick</i>

SKILLS

Languages: Java, JavaScript, Python, C#, C/C++, Go, Lua, HTML/CSS, SQL, MATLAB, JCL, COBOL, Easytrieve, PLI, REXX

Technologies: Azure DevOps, Jira, Git, Jenkins, Kubernetes, Docker, Linux, Bash, MySQL, PostgreSQL, Supabase, Firestore, Maven, Android Studio, Unity, Simulink, Qt, IDZ, Reflections, Node.js, React, NGINX, OpenAPI, OAuth2, GraphQL, Red Hat OpenShift, Helm, Figma, REAPER, Claude Code / AI-assisted development workflows

Research & methods: Mixed-methods user research (semi-structured interviews, diary studies, observational studies); quantitative analysis (ANOVA, regression, cluster and conjoint analysis, A/B testing); usability evaluation; Research through Design; physical prototyping (laser cutting, servo-actuated installations); startup business modelling, from market sizing and unit economics to commercialisation strategy

PROJECTS

StuddyBuddy, an AI-native academic planner

- Founded the product out of an entrepreneurial challenge at the University of Twente and carried it on past the course, now heading into closed beta.
- Validated the concept with a four-person team across 17 interviews and 3 observational studies, then built it on React and Supabase.

Vigil, EIT Digital Summer School venture

- Completed a two-week venture building intensive at the Polytechnic University of Bari, taking a concept from problem framing to Pitch Day in front of an industry and investor jury.
- Developed Vigil, a SaaS platform turning the mandatory EU Digital Product Passport from pure cost into revenue: AI closes data gaps and drafts supplier requests, while verified lifecycle data opens up second-hand market monetisation.

Pelle's Reflex Galleon, interactive installation

- Designed and built a live interactive experience as part of a project team, demoed publicly at a Dutch amusement park.
- Produced the character voice, projected animation and laser-cut components, and iterated the interaction design on visitor playtesting.

Human-centred design research project, workplace lactation rooms

- Ran human-centred research with a five-person team, carrying interviews and thematic analysis through ideation, prototyping and evaluation into a final design.

2D procedural map generator, BSc thesis

- Built a Unity tool in C# that generates layered 2D terrain from Perlin noise, cellular automata and flood fill, with live controls for map size, saturation, variation and minimum island size, plus map saving and loading.
- Grounded the build in a 10,000-word research paper surveying procedural content generation methods, which set the algorithm choices for the implementation.
- **Text Adventure Game:** GUI-based version of Zork built in C++ with Qt

Further projects and source code: github.com/richalex00

ACCOMPLISHMENTS

- Recipient of a full scholarship for the EIT Digital Master School
- A1 result for BSc thesis
- President's Volunteer Award Covid 19 for the Academic Year 2020/2021
- Recognition of Excellent Quality Credit Average (QCA) for the Spring Semester 2022/2023 (final semester, 4th year), QCA 3.68
- Recognition of Excellent Quality Credit Average (QCA) for the Spring Semester 19/20, QCA 3.6